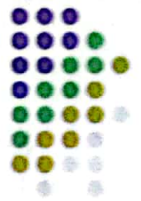


Alayen Iraqi University  
Engineering College/ AI Engineering Department



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3<sup>rd</sup> year

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Signals & Systems Lab

ذکار

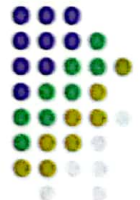
Lab ( 8 )

Systems

Lecturer: Amjed Jumaah



## Systems



$$y(t) = S\{x(t)\}$$

$$y[n] = S\{x[n]\}$$

a system is considered a *black box*

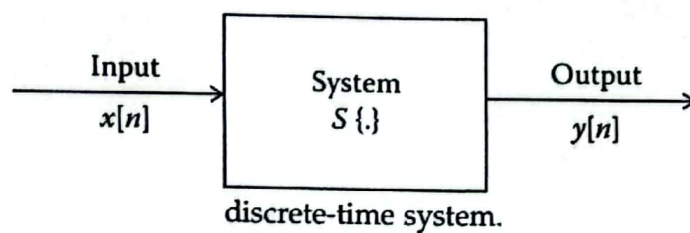
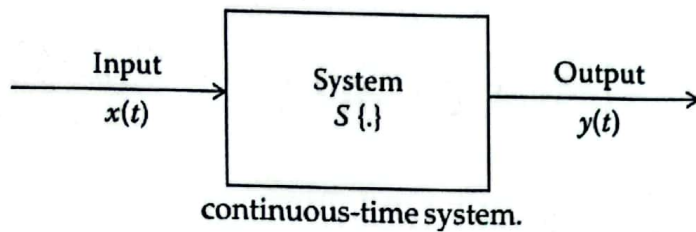
An example of a system is an electrical guitar.



### 3.1 Systems Classification

#### 3.1.1 Classification according to the Number of Inputs and Outputs

- Single-input single-output (SISO) systems



### 3.1 Systems Classification

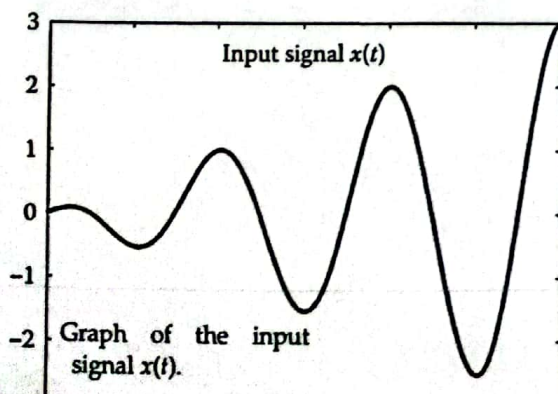
#### 3.1.1 Classification according to the Number of Inputs and Outputs

- Single-input single-output (SISO) systems

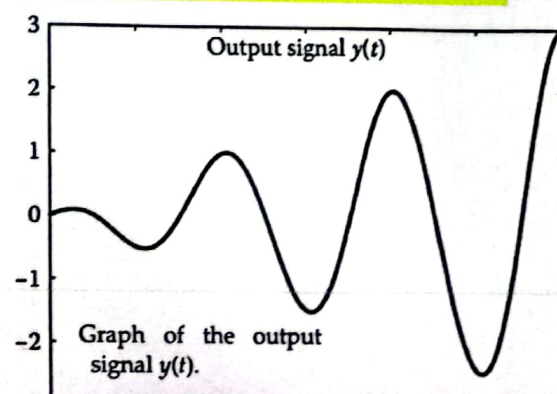
##### Example

The i/o relationship that describes a system  $S$  is  $y(t) = x(t - 2)$ . Compute and plot the system response to the input signal  $x(t) = t \cos(2\pi t)$ ,  $0 \leq t \leq 3$ .

```
t=0:.01:3;  
x=t.*cos(2*pi*t);  
plot(t,x);  
title('Input signal x(t)')
```



```
plot(t+2,x);  
title('Output signal y(t)')
```

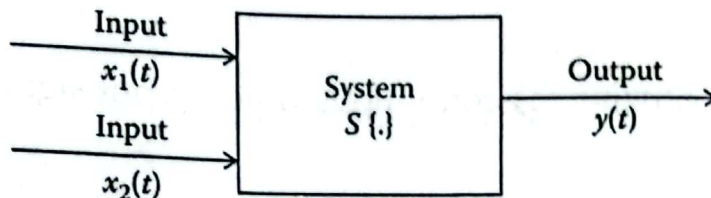




### 3.1 Systems Classification

#### 3.1.1 Classification according to the Number of Inputs and Outputs

- Multiple-input single-output (MISO)

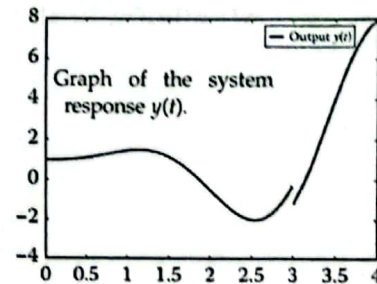
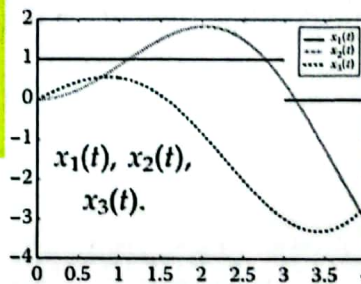


#### Example

Suppose that a MISO system  $S$  is described by the i/o relationship  $y(t) = x_1(t) + x_2(t) \cdot x_3(t)$ . Compute and plot the system's output if the input signals are given by  $x_1(t) = u(t) - u(t - 3)$ ,  $x_2(t) = t \sin(t)$ ,  $0 \leq t \leq 4$ , and  $x_3(t) = t \cos(t)$ ,  $0 \leq t \leq 4$ .

```
t=0:.01:4;
x1=heaviside(t)-heaviside(t-3);
x2=t.*sin(t);
x3=t.*cos(t);
plot(t,x1,t,x2,'-',t,x3,'-');
legend('x1(t)','x2(t)','x3(t)')
```

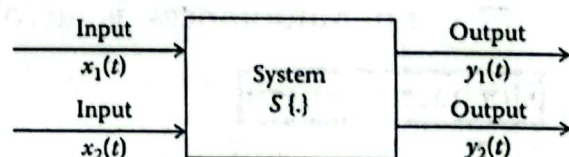
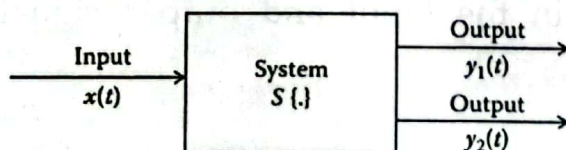
```
y=x1+x2.*x3;
plot(t,y);
legend('Output y(t)')
```



### 3.1 Systems Classification

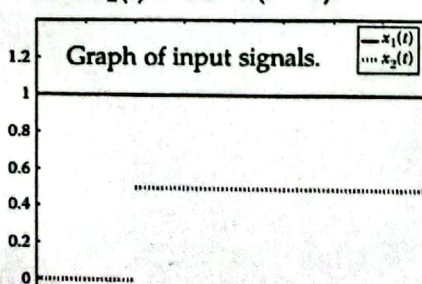
#### 3.1.1 Classification according to the Number of Inputs and Outputs

- Single-input multiple-output (SIMO)
- Multiple-input multiple-output (MIMO)



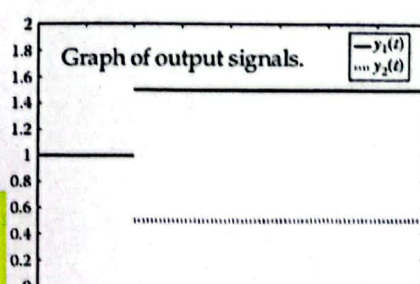
#### Example

Suppose that a MIMO system is described by the i/o relationships  $y_1(t) = x_1(t) + x_2(t)$  and  $y_2(t) = x_1(t) - x_2(t)$ . Compute and plot the system response to the input signals  $x_1(t) = u(t)$  and  $x_2(t) = 0.5 \cdot u(t - 1)$ .

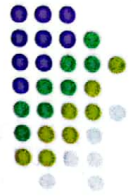


```
t=0:.01:4;
x1=heaviside(t);
x2=0.5*heaviside(t-1);
plot(t,x1,t,x2,'-');
ylim([-0.1 1.4]);
legend('x1(t)','x2(t)')
```

```
y1=x1+x2;
y2=x1-x2;
plot(t,y1,t,y2,'-');
ylim([-0.1 2]);
```



### 3.1.2 Continuous-Time and Discrete-Time Signals



“nature” of the input and output signals.

**a continuous-time system**

continuous-time input signals  $\Rightarrow$  continuous-time output signals

**discrete-time system**

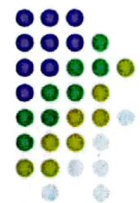
discrete-time input signal(s)  $\Rightarrow$  discrete-time output signal(s)

**hybrid systems**

An analog to digital (A/D) converter is an example of a hybrid system



### 3.1.3 Deterministic and Stochastic Systems



**deterministic system**

no randomness is involved in the input and output signals

**stochastic system**

does not always produce the same output for a given input.

